WS-04: Test Card TC-04 — Free Flight STOL Sortie Worksheet

FIRST FREE-FLIGHT SORTIE: TAXI, UNSTICK, CLIMB, CRUISE, FLARE & ROLLOUT

AIRCRAFT: MCO-1D Mark 1	**CARD ID:** TC-04	**DATE:** ____ / ____ / 2026	**LOCATION:** _____
OPERATOR / TECH: _____	**AIRFRAME S/N:** MCO-1D-001	**AMBIENT TEMP:** _____ °C	**BARO / QNH:** _____ hPa

1. Pre-Flight Electrical, Sensor & Environmental Go/No-Go Gate

System / Parameter	Inspection Requirement	Go/No-Go Safety Limit	Pre-Flight Measurement	Status
6S Main Motor LiPo Voltage	DMM resting check at battery lead	*≥ 25.0 V (≥ 4.17 V/cell)*	[ ] V	[ ] GO [ ] NO-GO
2S Receiver/Servo LiPo Voltage	DMM check at servo bus tap	*≥ 8.30 V (≥ 4.15 V/cell)*	[ ] V	[ ] GO [ ] NO-GO
Surface Wind Speed	Handheld anemometer on runway	*≤ 3.0 m/s aligned with runway*	[ ] m/s	[ ] GO [ ] NO-GO
Crosswind Component	Crosswind perpendicular to centerline	*≤ 1.0 m/s max crosswind*	[ ] m/s	[ ] GO [ ] NO-GO
Battery Sled Position Lock	Verify quick-pin in Hole 3 (250 mm)	*Hole 3 locked, Hole 6 pin in place*	[ ] VERIFIED HOLE 3	[ ] GO [ ] NO-GO
Pitot Cover Removed & Zeroed	Pixhawk telemetry indicated airspeed	*0.0 m/s ± 0.2 m/s calm*	[ ] m/s	[ ] GO [ ] NO-GO

2. Takeoff Acceleration & Climb Performance (Phase 1)

Flight Parameter	Telemetry Channel / Source	Design Predicted Target	Actual Flight Recorded	Tolerance / Target
Takeoff Ground Roll Distance	Runway visual markers / GPS	*≤ 12.0 m ground roll*	[ ] m	STOL target: 9–11 m
Unstick Indicated Airspeed ($V_{ ext{rot}}$)	Pitot airspeed at wheels-off	*6.20 m/s (13.9 mph)*	[ ] m/s	± 0.4 m/s
Elapsed Time to Unstick	Time from advance lever to liftoff	*3.5 to 4.5 seconds*	[ ] s	< 5.0 s limit
Steady Rate of Climb (V_z)	Pixhawk barometer / EKF3 climb rate	*1.80 to 2.20 m/s at 9 m/s*	[ ] m/s	Boyd P_s envelope
Flight Path Climb Angle (gamma)	gamma = arcsin(V_z / V_inf)	*11.5° to 15.0° climb*	[ ] °	Clearance obstacle
Peak Electrical Takeoff Power	6S bus current × voltage log	*≤ 680 W total peak*	[ ] W	< 32 A total

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CRUISE HANDLING, APPROACH, FLARE, TOUCHDOWN & POST-FLIGHT DEBRIEF

3. Level Cruise Flight & Handling Qualities (Phase 2)

Cruise Metric	Telemetry Source / Parameter	Nominal Design Target	Actual Flight Recorded	Performance Window
Level Cruise Airspeed (V_inf)	Pitot calibrated airspeed (CAS)	*9.00 m/s (20.1 mph)*	[ ] m/s	8.5 to 9.5 m/s
Active Cruise Flapping Frequency	ODrive commanded frequency	*2.20 to 2.50 Hz*	[ ] Hz	**Floor:** ≥ **1.87 Hz**
Cruise Throttle Lever Position	RC transmitter throttle channel	*48% to 55% detent*	[ ] %	Cruise detent
Mean Electrical Power Consumption	Average bus power in steady level	*240 to 280 W*	[ ] W	< 300 W limit
Specific Energy (Cost of Transport)	Wh energy consumed per km	*7.4 to 8.6 Wh/km*	[ ] Wh/km	< 9.5 Wh/km
Pitch Trim Stick Bias	Elevator trim offset for level flight	*0.0% ± 5.0%*	[ ] %	Validates 26.5% CG
Roll Trim Stick Bias	Aileron trim offset for straight track	*0.0% ± 3.0%*	[ ] %	Validates wing balance

4. Approach, Flare, Touchdown & Rollout (Phase 3)

Landing Metric	Target Expected	Actual Flight Recorded	Safety Evaluation & Notes
Final Approach Indicated Airspeed	*7.50 m/s ± 0.3 m/s*	[ ] m/s	Stabilized glideslope approach
Flare Airbrake Command Response	*Clamshell high-drag flare*	[ ] CLEAN FLARE [ ] SLOW	H-54 / CONV-BRK deceleration
Touchdown Airspeed (Wheels-On)	*4.50 to 5.20 m/s*	[ ] m/s	Gentle wheel contact
Landing Ground Rollout Distance	*≤ 8.5 m from touchdown to stop*	[ ] m	Du-Bro wheel rollout braking
Peak Vertical Touchdown Load	*≤ 1.8 g vertical spike*	[ ] g	Gear damping performance

5. Post-Flight Battery Recovery & Thermal Health Inspection

Flight Duration: [ ] min : [ ] sec	**6S Rested Voltage: [ ] V (≥ 21.8V)**	**Capacity Consumed: [ ] mAh**
2S Rested Voltage: [ ] V (≥ 7.4V)	**Max Motor Temp: [ ] °C (≤ 65°C)**	**Max ODrive FET Temp: [ ] °C (≤ 75°C)**
Dyneema Lines Post-Flight: [ ] ZERO FRAYING [ ] WITNESS MARKS INTACT [ ] TENSION VERIFIED (180N)		

GATE CERTIFICATION: TC-04 FREE FLIGHT ENVELOPE EXPANSION CERTIFICATION	**STATUS: [ ] PASS [ ] CONDITIONAL [ ] FAIL / RE-TEST**
INSPECTOR / LEAD: _____	**SIGNATURE & DATE: _____**
REMARKS / DISCREPANCIES: _____	